

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – OCTOBER, 2008.

Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer book itself.
5. Maximum time allowed is THREE hours.

1. A Predator beast weighs 2008kg at the beginning of a year. During the first month of the year its weight is increased by $33\frac{1}{3}\%$ and in the second month decreased by 25% and in the third month increased by 50%, fourth month decreased by $33\frac{1}{3}\%$, fifth month increased by $16\frac{2}{3}\%$ and in the sixth month decreased by $21\frac{3}{7}\%$. This increase and decrease of the weight continued in the next 6 months in the same order with the same percentage. Find the weight of the predator at the end of the year.
2. ' n ' is a natural number greater than 1. Prove that $(n^8 + n^7 + 1)$ is always composite.
3. $ABCD$ is a trapezium, such that $AB \parallel CD$. $AB = 10\text{cm}$; $BC = 8\text{cm}$; $CD = DA = 5\text{cm}$. Find the height of the trapezium and also find AC and BD .
4. Three regular polygons have one vertex in common and just fill the whole space at that vertex. If the number of sides of the polygons are a , b and c , prove that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{2}$.
5. Find the number of all 5-digit numbers such that their squares have 9 digits. What is the largest of these numbers? [observe: $3^2 = 9$; $4^2 = 16$; $31^2 = 961$; $32^2 = 1024$]
6. A certain 8 digit number is divided by 11 and the remainder is r , $0 < r < 11$. The digits are reversed and divided by 11. Find the remainder in terms of r .
7. Given l , m , n are reals and $l + m \neq 0$, $n + l \neq 0$, $m + n \neq 0$ and $\left(\frac{m}{m+n}\right)^2 + \left(\frac{n}{n+l}\right)^2 + \left(\frac{l}{l+m}\right)^2 = \left(\frac{n}{m+n}\right)^2 + \left(\frac{l}{n+l}\right)^2 + \left(\frac{m}{l+m}\right)^2$. Prove that two of the three quantities l, m, n must be equal.

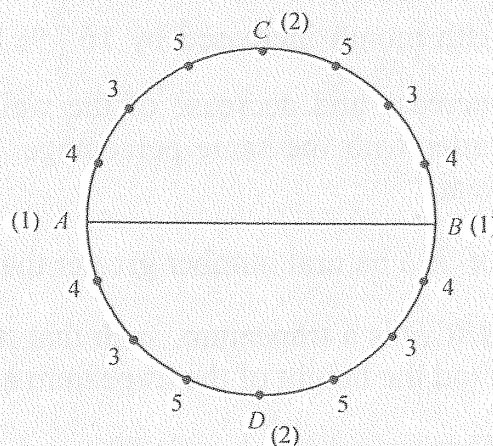
8. A natural number is said to be a "super-star-number" if the number is less than 10 times its product of the digits. Find how many super-star-numbers are there from 10 to 200.
9. ABCD is a convex pentagon with $\angle A = \angle B = \angle D = 90^\circ$; $\angle C = \angle E$. Sides AB, BC, CD, DE and EA are extended to K, L, M, N and O respectively. The exterior angles $\angle OAB$, $\angle KBC$, $\angle LCD$, $\angle MDE$ and $\angle NEA$ are bisected and the bisectors are produced in either direction to form the pentagon PQRST. Find the angles of the pentagon.
10. Among all the rectangles of area 2008 sq. units, what is the smallest possible value of the sum of the lengths of the diagonals? What is the ratio of the product of the diagonals and the area of such a rectangle? Compare this ratio with the ratio of the product of diagonals of the square to that of the area of the square.

11. In the adjoining figure, AB = 200 units is a diameter of the circle. Points A, B are given the numbers 1,1. The two semicircles are bisected at C,D. Numbers 2,2 are given to C,D (see figure).

Each of the quarter circles are bisected and given the numbers 3,3,3,3 (see figure)

Now, each arc is bisected and the point is given the number which is the sum the two numbers at the end points of the respective arcs (see figure)

This process is continued till the sum of all numbers on the circle is atleast numerically equal to the product of four times the area of the circle and one third the radius. Find the least number of points plotted on the circle. (Take $\pi = 3.14$)



12. If a, b, c are reals and $abc = 1$ and $a + b + c = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$, prove that one of these real numbers is equal to 1.